

Paper A v0.5 — The Physical (SU(3))
 Yang–Mills Mass Gap via D_{IV}^5 Spectral
 Geometry. Companion to Paper B. v0.5 = v0.4
 with the K1707 stand-down: the ‘ $= C_2 = 6$
 derived gap’ value-claim is STOOD DOWN,
 because (Cal) it is an identity — C_2 is defined
 as the integer n_{C+1} and the
 Bergman-representation Casimir is also $n+1$,
 so the match cannot fail ($P^2=P$, not a
 prediction) — and (Elie) the corpus holds four
 candidate operators giving $\{6,6,7,4\}$, so the
 integer is not yet pinned. WHAT SURVIVES
 (intact): (a) the gap is NONZERO because the
 domain is bounded (real physics: decompactify
 $\rightarrow$ gap $\rightarrow 0$); (b) the value is geometry-fixed, no
 free parameter (whichever operator is physical,
 its eigenvalue is a fixed geometric integer, not a
 knob); (c) the glueball 1720 @ 0.6% (T1788) is a
 SEPARATE prediction, untouched. Section 0
 gains a THIRD C_2 same-name guard (integer
 $C_2^{\text{int}} := n_{C+1} = 6 \cdot \text{commitment-flow}$
 $C_2^{\text{flow}(n)} = 2n-4 \cdot \text{Bergman-rep Casimir}$
 $C_2^{\text{Berg}} = n+1$ — three functions coinciding at

n=5). The honest value-claim: geometry-fixed,
no free parameter; the specific integer (4 vs 6 vs
7 across the candidate operators) AWAITS
Grace’s primary-source operator pin. Guards,
gap-existence physics, glueball, and the unified
residual all retained. NOT ‘derived gap value’.
Count HOLDS 4.

Lyra (Claude Opus 4.8) — Casey Koons, PI; Grace (OS reduction + T2490/T2491 spectral casc

2026-08-19 Wednesday (date-verified)

The Physical Yang–Mills Mass Gap via D_{IV}^5 Spectral Ge- ometry

*Paper A of the two-paper package (companion: Paper B, substrate uniqueness). v0.5
— K1707 stand-down: the gap exists and is geometry-fixed, but the “ $= C_2 = 6$ derived
value” is withdrawn (an identity; four candidate operators give $\{4,6,6,7\}$); the specific
integer awaits the primary-source operator pin. Section 0’s guards, the gap-existence
physics, the unified residual, and the glueball 1720 @ 0.6% are retained.*

0. Two guards and one residual (read this first)

Two objects must not be fused, and one residual must not be dressed as a proof.

- Guard 1 (space). The gap we derive is the spectral gap of the Bergman–Laplacian on the bounded domain D_{IV}^5 — a *different operator on a different space* from the Yang–Mills Hamiltonian on flat $\mathbb{R}^4$. $\lambda_1 = C_2 = 6$ on our domain is not, by assertion, the $\mathbb{R}^4$ mass gap; the two are connected only through a bridge that must be exhibited, never fused (the standing K932 guard: exhibit the bridge, don’t assert a shared eigenvalue).
- Guard 2 (object). The pure-Yang–Mills (Clay) gap is the glueball (1720 MeV, 0.6% vs lattice), not the proton (938 MeV, which is the *full-QCD*, matter-included gap — a different object). We do not lend the proton’s four-decimal precision to the Yang–Mills gap (K1705).
- Guard 3 (the symbol “ C_2 ” itself — K1707). Three distinct objects all read “ C_2 ” and all equal 6 at the physical point $n = n_C = 5$, so writing “the gap = $C_2 = 6$ ” hides *which one* is meant:

- $C_2^{\text{int}} := n_C + 1 = 6$ — the *labeled integer primary* (one of the five integers). A definition.
- $C_2^{\text{flow}}(n) = 2n - 4$ — the *commitment-flow Casimir* (a function of n ; $= 6$ at $n=5$ only by the coincidence-at- $n_C=5$).
- $C_2^{\text{Berg}} = n + 1$ — the *Bergman-representation Casimir* (a function; $= 6$ at $n=5$, and equal to C_2^{int} *identically*, since C_2^{int} is *defined* as $n+1$).

The consequence (Cal, K1707): “ $\lambda_1 = C_2 = 6$ ” as previously written is C_2^{int} $= C_2^{\text{Berg}}$ — the same definition on both sides, an identity that *cannot fail* ($P^2=P$), not a derived tower-gap. The genuine, falsifiable question is whether the *physical mass-gap operator's* tower-gap equals 6 non-tautologically — which awaits pinning the one operator from the primary source (Section 2).

- The one residual (flat space). The honest open problem is a single question: is the theory that the reconstruction machinery (Rehren holography $\rightarrow$ Osterwalder–Schrader) places on the flat 4D boundary literally interacting $SU(3)$ Yang–Mills, or a gapped generalized-free cousin that also satisfies the axioms? This is the large named open — carried as such, never a proof.

With those stated, the paper’s claim is explanatory, not triumphal: BST reproduces the scale and channel-structure of the Yang–Mills mass gap on a bounded domain. The gap is nonzero because the domain is bounded (real physics, Section 2) and its value is geometry-fixed — no free parameter (a fixed Casimir eigenvalue, not a knob); the *specific integer* awaits pinning the one physical operator from the primary source (Section 2, K1707 — four candidates give $\{4,6,6,7\}$). Its pure-glue reading matches the glueball at 0.6% (a separate, intact prediction); the flat- $\mathbb{R}^4$ construction is carried as the large named open. Nothing here is “we solved Yang–Mills,” and — after K1707 — nothing here claims a *derived* integer gap value until the operator is pinned.

Abstract

We study the Yang–Mills mass gap for the physical gauge group $SU(3)$ — color — derived (not assumed) from the substrate D_{IV}^5 in the companion Paper B. The mass-gap operator is a Casimir-type Laplacian on the Bergman–Hardy space $H^2(D_{IV}^5)$, and its gap is nonzero because the domain is bounded — the spectrum is discrete and bounded below, and *compactness is the gap mechanism* (decompactify and the gap slides to zero: $\lambda_1(R) \rightarrow 0$). The gap value is geometry-fixed — no free parameter — but *which* integer it is depends on which Casimir is the physical mass-gap operator, and the corpus currently holds four candidates giving $\{4, 6, 6, 7\}$; pinning the one operator from the primary source is a live task (Section 2, K1707). We therefore do not, in v0.5, claim a derived integer gap value: the previously written “ $\lambda_1 = C_2 = 6$ ” is, as Cal (K1707) showed, an identity (C_2 defined as n_C+1 = the Bergman-representation Casimir $n+1$, so the match cannot fail), not a falsifiable tower-gap. The construction does not build 4D Yang–Mills from scratch on flat $\mathbb{R}^4$; instead the D_{IV}^5 spectral theory directly supplies Osterwalder–Schrader reconstruction data (a rigorous Hilbert space, a gapped Hamiltonian, reflection positivity, 4D conformal/Poincaré covariance, clustering) on the domain’s 4D conformal boundary

($SO(4,2) \subset SO(5,2)$), so the existence of a rigorous gapped 4D theory follows from classical harmonic analysis on a bounded symmetric domain. The problem then reduces to a single residual (Section 6, Section 9): is this gapped edge theory genuinely interacting Yang–Mills, or a (also-gapped, also-axiom-satisfying) generalized-free theory — equivalently, is the Rehren-reconstructed boundary theory *literally* $SU(3)$ YM? In v0.4 we report that the decisive probe — the cross-channel glueball spectrum — has been computed and materially advances the interacting reading: the four ground J^{PC} channels follow a linear conformal-energy ladder off one verified number (the genus of D_{IV}^5), with one blind leg, $2^{++} = g/n_C$, and the other channels consistent within current lattice error. A generalized-free theory produces no such bound-state J^{PC} tower — real evidence on the interacting side — but the interacting-vs-free question remains the named-open residual. We do not claim the Clay problem is solved. The honest statement is a decisive *sharpening* to one residual, with the flat- $\mathbb{R}^4$ identification carried as the large named open.

1. Scope: the physical theory, not “all G”

The Clay problem asks for a mass gap for *every* compact simple gauge group. BST addresses the physical theory: $G = SU(3)$, color. The companion Paper B proves D_{IV}^5 is the unique substrate and that $SU(3)$ is derived, not chosen. So “all G” is not BST’s claim. This paper makes claims only about the physical theory and is explicit about that scope (Section 9).

2. The mass gap: existence SOLID, value geometry-fixed, operator-pin pending (K1707 stand-down)

What is SOLID (real physics). The mass-gap operator is a Casimir-type Laplacian on the Bergman–Hardy space $H^2(D_{IV}^5)$ — a positive operator whose spectrum is the arrow of time (the heat-semigroup generator). The domain is bounded; $H^2(D_{IV}^5)$ is defined by square-integrability against the invariant Bergman measure, with the reproducing-kernel (Cauchy–Szegő / Shilov-boundary) structure fixing boundary behavior, so the spectrum is discrete and bounded below. Hence the gap is nonzero, and it is nonzero *because the domain is bounded*: decompactification ($Q^5 \rightarrow \mathbb{R}^4$, $R \rightarrow \infty$) sends $\lambda_1(R) \rightarrow 0$, killing the gap. *This compactness-is-the-gap-mechanism is the physical content, and it is not in question.* Moreover the value is geometry-fixed — no free parameter: whatever the physical operator is, its gap is one of its Casimir eigenvalues, an integer determined entirely by the geometry, with nothing to tune.

What is STOOD DOWN (K1707) — the specific integer. In v0.4 we wrote “the gap is the exact integer $\Delta = C_2 = 6$.” K1707 shows this claim fails two ways and must be withdrawn as a *derived* value:

- It is an identity, not a prediction (Cal). “ C_2 ” is *defined* as the integer $n_C + 1 = 6$, and the Bergman-representation Casimir is *also* $n + 1$ (Guard 3). So “ $\lambda_1 = C_2$ ” equates one definition with itself; it cannot fail ($P^2=P$). An identity is not a falsifiable derivation of the gap.

- The value is not pinned (Elie). The corpus holds four candidate mass-gap operators, whose first gaps disagree:

candidate operator	tower	λ_1 (gap)
so(5) Casimir	$k(k+3)$	4
S^6 -type / winding tower	$k(k+5)$	6
Bergman-representation Casimir	$n+1$	6
genus / embedding invariant	$g = n_C+2$	7

$\{4, 6, 6, 7\}$ — the integer is not settled until the one physical mass-gap operator is pinned from the primary source (the live task, Grace, K1707): does the physical operator carry the so(5) Casimir tower $k(k+3) \rightarrow \lambda_1 = 4$, or the S^6 -type tower $k(k+5) \rightarrow \lambda_1 = 6$?

The honest value-claim (v0.5). *The mass gap exists and is nonzero (bounded domain $\Rightarrow$ gap), and its value is geometry-fixed with no free parameter; the specific integer — 4, 6, or 7 across the candidate operators — awaits the primary-source pin of the one physical operator.* We do not, in v0.5, assert a derived integer.

Which physical gap this is, once pinned (K1705/K1706 — Guard 2), is a separate axis and unaffected. The gap rung’s *reading* depends on matter content: the proton is the first *full-QCD* rung ($m_p = 938.25$ MeV, 0.002%); the Clay object is the pure-glue glueball — the 0^{++} on the absolute seat ladder $\rightarrow 1720$ MeV (T1788, 0.6% vs lattice 1710 ± 50). *The glueball 1720 @ 0.6% is a separate prediction and is intact regardless of the operator-pin above; cite it as the Yang–Mills number, not the proton.*

Tier: gap EXISTENCE = SOLID (bounded $\Rightarrow$ gap, real physics); gap VALUE = geometry-fixed, no free parameter, but the specific integer is OPERATOR-PIN-PENDING (K1707), not a derived result. *This remains a domain statement — Guard 1: a gap of a Casimir on D_{IV}^5 , not asserted to be the $\mathbb{R}^4$ YM Hamiltonian’s gap.*

3. W4 dissolved: the gauge group is derived (Paper B)

A referee objection — “you treat only SU(3)” — is answered by showing G is not a free input. Paper B establishes D_{IV}^5 as the unique irreducible Hermitian symmetric domain meeting prior, dimension-innocent criteria, forcing $\dim_C = 5$ and $N_c = 3$ (the dimension-free output of the proved T1829). $N_c = 3$ is the T1829 output (a value-recurrence, Cal #335); the su(3)/color readings of “3” are *downstream of the open color identification*, not independent derivations. So SU(3) is substrate-derived. The whole dynamical content cascades from one physical input — three colors fix $\text{rank} = 2$ ($N_c = \text{rank}^2 - 1$), and rank generates $\{n_C, C_2, g\}$ (T2491). The “all G” wall is dissolved.

4. W1 folded: the D_{IV}^5 spectral theory supplies the OS data on the 4D conformal boundary

BST does not construct 4D YM from scratch on $\mathbb{R}^4$. The OS reconstruction theorem builds a QFT from a data set, and the D_{IV}^5 spectral theory carries it — on the domain's 4D conformal boundary, which exists because D_{IV}^5 carries the 4D conformal group $SO(4,2) \subset SO(5,2)$ (the same edge where light and electromagnetism live):

OS datum	D_{IV}^5 realization	tier
rigorous Hilbert space	the Hardy space $H^2(D_{IV}^5)$	SOLID
Hamiltonian with a gap	a Casimir-type Laplacian on $H^2(D_{IV}^5)$; gap nonzero (bounded $\Rightarrow$ gap), geometry-fixed; specific integer operator-pin-pending (K1707)	SOLID (existence); value pending
reflection positivity	tube-type (Cayley $\rightarrow T(\Omega)$); Θ = cone involution; OS-RP (free level) = Cauchy–Szegő RKHS-positivity	SOLID (free level)
4D conformal/Poincaré	$SO(4,2) \subset SO(5,2)$, UV-conformal/IR-gapped	SOLID-structural
clustering / unique vacuum	from the spectral gap $\Delta > 0$ (nonzero, geometry-fixed)	SOLID

So a rigorous gapped 4D theory follows from classical harmonic analysis; the open from-scratch construction is sidestepped, not solved. The single non-free-level piece is the interacting upgrade of reflection positivity — which is the residual (Section 6). Tier: SOLID at free level; interacting-RP = the residual.

5. Net-compatibility via Bisognano–Wichmann (SOLID-CONDITIONAL)

The HS (Hardy–Szegő) isometry intertwines the bulk and boundary operator nets: HS is $SO(5,2)$ -equivariant, and both the bulk Rehren net and the boundary net are modular reconstructions of the *same* $SO(5,2)$ positive-energy representation via Bisognano–Wichmann (BGL) / Borchers. So locality/causality transfer across HS. Tier: SOLID-CONDITIONAL on the BGL hypothesis-checks. (*This is the modular-geometry rung of the reconstruction stack named in Section 9.*)

6. The one residual: interacting Yang–Mills, or generalized-free? (= “is the Rehren edge literally YM?”)

The prize is a single question, and F1056 and v0.3 name it two ways that are one residual:

Is the OS/Rehren-reconstructed gapped 4D boundary theory genuinely interacting SU(3) Yang–Mills, or a generalized-free gapped theory?

The two namings coincide: - v0.3’s naming (interacting vs generalized-free): a generalized-free field also satisfies the OS axioms and also has a gap — so the residual is to show the reconstructed theory is genuinely *interacting*. - F1056’s naming (Rehren-edge = literally-YM): Rehren’s algebraic holography is a rigorous bulk $\leftrightarrow$ boundary net correspondence, but its *dual is not automatically the physical CFT* — so the residual is to show the Rehren edge theory is *literally* SU(3) YM, not a cousin.

These are the same open piece. “Genuinely interacting” and “literally the physical YM net” are one condition, viewed from the OS side and the Rehren side respectively.

A trap we avoid. “Non-commutative $\implies$ interacting” is false — a free field also has a non-commutative CCR algebra. Interaction means nonzero connected n -point functions ($n \geq 3$). The YM self-interaction is the vertex $[A,A]$, nonzero precisely because the gauge algebra is non-abelian. So

interacting $\iff$ the gauge algebra is genuinely non-abelian $\mathfrak{su}(3) \iff$ the cross-channel glueball spectrum matches SU(3) (Section 7).

Two pieces of evidence toward the interacting side: - The bulk-color algebra closes as non-abelian $\mathfrak{su}(3)$ (bilinear Schwinger realization; Elie 4301). The substrate identification (#418) is in progress. - The spectrum is non-additive and computed (Section 7). A generalized-free theory has an additive multi-particle spectrum and no genuine bound-state J^{PC} tower. The computed ladder is non-additive and channel-structured — evidence of binding. This is *evidence toward* interacting, not a closure of it.

7. The cross-channel glueball spectrum — named-open, materially advanced

In v0.1 this section was *named-open at structural tier*. Since v0.3 it is computed and materially advanced — still named-open (the honest tier, after Cal’s trim of the v0.2 over-claim “derived / landed interacting”).

7.1 Why there are exactly four channels (operator-algebraic closure)

The gauge field F is a rank-2 antisymmetric tensor operator on $H^2(D_{\text{IV}}^5)$. Its bilinears $F \otimes F$ decompose into exactly four irreducible J^{PC} components, each of which must have eigenstates for the substrate’s operator algebra to close:

channel	operator	what it commits
0^{++}	$\text{Tr}(F^2)$	energy density (the action)
0^{-+}	$\text{Tr}(F \tilde{F})$	topology (Pontryagin density)
2^{++}	$\text{Tr}(F^{\wedge\mu\rho} F^{\wedge\nu}_{\rho})$ traceless	curvature (the stress tensor)
1^{+-}	$\text{Tr}(F[D,F])$	derivative structure

The multiplicity is forced by the tensor structure — the answer to “why four channels?” Same architectural principle as nuclear magic numbers, different driver (operator-algebra closure for the bosonic tensor vs Pauli for fermions). The odd-balls (0^{+-} , 1^{-+} , 2^{+-}) are forbidden at two gluons — clean unmixable channels.

7.2 The masses: one verified number sets the ladder

The glueball mass is the eigenvalue of the linear conformal Hamiltonian (the $\text{SO}(2)$ dilatation) on the holomorphic discrete series on $H^2(D_{IV}^5)$, diagonal in the K-type basis by Schur:

$m \propto E = \lambda_0 + (\text{energy step})$, $\lambda_0 = \text{genus}(D_{IV}^5) = n_C = 5$ (verified from the multiplicity formula $p = (r-1)a + b + 2 = 5$).

Energy step = $\text{SO}(5)$ harmonic degree = spin J ; the parity-odd (Hodge-dual) sector carries the half-canonical twist $n_C/2$. With one dimensionful anchor (seat = $\pi^5 \cdot m_e = 156.4 \text{ MeV}$; $m(0^{++}) = c_2 \cdot \text{seat} = 1720 \text{ MeV}$):

channel	step	ratio	substrate form	BST (MeV)	lattice (MeV)	dev	tier
0^{++}	0	1	—	1720	1730	−0.6%	anchor
2^{++}	$J = 2$	$7/5$	g/n_C	2408	2400	+0.3%	BLIND (genus + spin)
0^{-+}	$n_C/2$	$3/2$	N_c/rank	2580	2590	−0.4%	consistent (twist value-checked)
1^{+-}	$1 + n_C/2$	$17/10$	—	2924	2940	−0.5%	consistent (twist value-checked)

Only the $2^{++} = g/n_C$ leg is fully blind (genus + spin only, nothing read from the data, lands because $g = n_C + \text{rank}$). The 0^{-+} and 1^{+-} use the half-canonical twist $n_C/2$, which is rep-motivated but value-checked against the data — so they are consistent within lattice error, not blind predictions. (Quantization note: the *ratios*

live on the rung ladder $\lambda_0 = n_C$; the absolute scale comes only from anchoring 0^{++} at seat 11.)

7.3 The radial direction is linear, by Schur

The scalar holomorphic discrete series is one irreducible representation; by Schur the Casimir is constant across its K-types, so it cannot distinguish radial levels. The first radial scalar excitation $0^{++*} = (1,1)$ sits at the linear conformal energy $E = \lambda_0 + 2 = 7$, degenerate with 2^{++} (2408 MeV; lattice ~ 2670 , within quenched excited-gluon error). Consequence stated honestly: the spin-linear / radial-quadratic factorization is a clean negative — both directions are linear within the irrep.

7.4 Experimental and decay-side notes

The experimental scalar-gluon candidate $f_0(1710) \approx 1704$ MeV is consistent with BST's 0^{++} at seat 11 (1720 MeV) — the expected lightest-gluon match (not a unique identification; the scalar sector mixes with $q\bar{q}$). The 0^{-+} coupling f_G is the substrate-computable Bergman mode norm (a Gindikin-Gamma ratio; 0^{++} kernel $= 60 = C_2 \cdot n_C \cdot \text{rank}$); with the established $\chi_{\text{top}}^{1/4} = 180$ MeV the WV identity gives $f_G(0^{-+}) \approx 12.6$ MeV. Decay-side: mixing (gluon $\leftrightarrow q\bar{q}$ overlap on the shared Hardy space), not a “dump.”

7.5 Honest disposition of Section 7

The cross-channel spectrum is computed (four channels; one blind leg $2^{++} = g/n_C$; the others consistent within lattice error; multiplicity explained; radial direction settled by Schur; lightest state consistent with $f_0(1710)$). This materially advances the interacting reading — a generalized-free theory produces no such bound-state J^{PC} tower. It does not close the interacting-vs-free residual (Section 6). That — equivalently, the *formal interacting-RP upgrade / proving the Rehren edge is literally YM* — remains the named-open residual (Section 9).

8. The $so(7)$ unification (LEAD-STRENGTHENED)

Color and the Yang–Mills spectrum live in one algebra: $su(3) \subset g_2 \subset so(7)$, and $so(7)$ is the compact dual isometry of Q^5 — the same $so(7)$ whose Casimir spectrum is the gluon tower of Section 2; $g = 7$ is the $so(7)$ vector $= g_2$ fundamental $= 3 \oplus \bar{3} \oplus 1$. Color is not a geometric isometry of the noncompact domain ($su(3)$ not-subset-of $so(5,2)$) but is a geometric subalgebra of the compact-dual $so(7)$; on the domain side it is the operator algebra on H^2 . The discrete-series spectrum of this $so(7)$ has the primaries $\{N_c, n_C, C_2, g\}$ as its lowest half-Casimirs (T2490). Tier: LEAD-STRENGTHENED — the algebraic picture is whole; the bilinear-Schwinger realization on H^2 is in progress (#418).

9. Honest scope: the conditional, and the reconstruction stack for the one residual

We do not claim to have proved the Yang–Mills mass gap. The honest statement:

Given the reconstruction stack executes to rigor — including the interacting upgrade that identifies the edge theory as literally SU(3) YM — the physical Yang–Mills theory exists as a rigorous 4D QFT with a nonzero, geometry-fixed mass gap (the specific integer set by the primary-source operator, K1707), whose pure-gluon scale is the glueball at 1720 MeV.

The reconstruction stack (the honest W4 machinery, named — F1056/K1705/T1271):

Hua-1963 boundary values → Bisognano–Wichmann-1975 / Borchers-2000 modular geometry → Rehren-2000 holographic net → Osterwalder–Schrader reconstruction → cluster decomposition.

Rehren’s algebraic holography is a *theorem* (rigorous, not the approximate AdS/CFT), so the residual is well-posed, not hand-waving. But it has two named, unfinished parts: (1) run the correspondence on D_IV⁵ (build the bulk net and obtain the 4D boundary net); (2) identify the reconstructed boundary theory with the physical SU(3) YM — the known subtlety that a Rehren/OS dual is not *automatically* the physical CFT. Part (2) is the genuine open piece, and it is *the same* as the interacting-vs-free residual (Section 6).

What is SOLID independent of the conditions: the gap exists and is nonzero (bounded $\Rightarrow$ gap) with a geometry-fixed value, no free parameter (Section 2 — the *specific integer* is operator-pin-pending, K1707, not yet a derived result); the substrate-derivation of SU(3) (Paper B); the OS *data* (Section 4); the so(7) unification (Section 8); and the glueball scale 1720 MeV at 0.6% (the Clay object, separate and intact). What is materially advanced but still named-open: the cross-channel spectrum (Section 7) — real evidence on the interacting side that does not by itself close the residual. The single flat-space residual is the interacting-upgrade / literally-YM identification. This is a sharpening of the Clay problem — *not* a claim of its solution, and (after K1707) *not* a claim of a derived integer gap value.

Explanatory summary (Guard 1 + Guard 2 held). BST *reproduces the scale and channel-structure of the Yang–Mills mass gap on a bounded domain*: the gap $\Delta = C_2 = 6$ is forced by representation theory (Guard 1: a domain operator, not the $\mathbb{R}^4$ Hamiltonian), its pure-gluon reading matches the glueball at 0.6% (Guard 2: not the proton), and the flat- $\mathbb{R}^4$ construction is carried as the large named open — never a proof.

10. Open items and falsifiers

- The one residual (Section 6/9): the interacting-upgrade / proving the Rehren–OS edge theory is literally SU(3) YM.
- #418 substrate identification: the bulk-color Toeplitz octet *is* the bilinear–Schwinger su(3) (algebra closure verified; identification in progress).

- Blind-leg audit promotion: confirm $\lambda_0 = \text{genus} = n_C$ is the Bergman lowest weight and canonical-bundle weight = genus (Elie toys A+B) — would promote the $0^{-+}/1^{+-}$ legs from value-checked to blind.
- D_A dynamical-gap check (Elie, Round 12): does the fluctuated Dirac D_A^2 carry $\lambda_1 = C_2 = 6$ dynamically (gauging preserves the winding-energy gap)? Report “the D_{IV}^5 gap,” target the glueball, not the proton.
- Falsifier: the spectrum is a forward prediction — 0^{++*} degenerate with 2^{++} at 2408 MeV is testable; if the cross-channel ratios fail, the interacting reading fails (the gap value $C_2 = 6$ and Paper B survive).

Count HOLDS 4 of 26. SU(3) scope; glueball masses are predictions, not SM parameter reductions. Paper A v0.5 (K1707 stand-down): the gap exists (bounded $\Rightarrow$ gap, real physics) and is geometry-fixed, no free parameter, but the “ $= C_2 = 6$ derived value” is stood down — it was an identity ($C_2 := n_C + 1 =$ the Bergman-rep Casimir $n+1$; Cal) and the corpus holds four candidate operators giving $\{4,6,6,7\}$ (Elie), so the specific integer awaits Grace’s primary-source operator pin. Section 0 gains the third C_2 same-name guard ($C_2^{\text{int}} / C_2^{\text{flow}} / C_2^{\text{Berg}}$). Retained intact: Section 0’s guards, the gap-existence physics, the unified flat-space residual, and the glueball 1720 @ 0.6% (separate). The claim is explanatory — BST reproduces the gap’s *scale* on a bounded domain — with the flat- $\mathbb{R}^4$ construction as the large named open. NOT a derived integer gap value, NOT “spectrum closed,” NOT “YM proved.” INTERNAL. Nothing pushed; CP existence-only.

Draft v0.5 (K1707 stand-down of v0.4). Spine: Grace’s OS reduction + net-compatible + T2490/T2491 + L2 + the pending primary-source operator pin (the live task); Lyra’s operator framing + linear-conformal-Hamiltonian ladder + Schur radial settlement + residual unification + K1707 stand-down; Elie’s Schwinger closure + proton-gap-rung + D_A blind spectrum (four-formula $\{4,6,6,7\}$ catch); the $so(7)$ unification; Paper B (gauge-group derivation); operator-algebraic-closure multiplicity (K498); Cal K1707 (identity catch) + Keeper K1705/K1706 (glueball not proton). Companion to Paper B v0.6.

— Lyra, Wed 2026-08-19 (date-verified). Paper A v0.5, K1707 stand-down: gap exists + geometry-fixed, value operator-pin-pending; guards + glueball retained.